

* Hamiltonians

$$i\hbar \frac{\partial \psi(r,t)}{\partial t} = \hat{H} \psi(r,t)$$

$$\hat{H} = \frac{\hat{p}^2}{2m} + \hat{V}(r) = \frac{\hbar^2}{2m} \nabla^2 + \hat{V}(r)$$

↓
Hamiltonian

where $\nabla^2 = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2} \rightarrow \text{Laplacian Operator}$

For $\hat{H} = \hat{H}(\hat{x}, \hat{p}) \rightarrow \hat{H}$ is constant with time

$$\psi(r,t) = e^{-i\hat{H}t/\hbar} \psi(r,0)$$

For $\hat{H} \psi(r,0) = E \cdot \psi(r,0)$

$$\psi(r,t) = e^{-iEt/\hbar} \cdot \psi(r,0)$$

$E \rightarrow \text{Energy} \rightarrow \text{Eigenvalue} \rightarrow \text{Real}$

$\hat{H} = \hat{H}^\dagger \rightarrow \text{Hermitian matrices have only real eigenvalues}$

$$U(t) = e^{-i\hat{H}t/\hbar} \rightarrow \hat{H} = \hat{H}^\dagger \rightarrow U^\dagger U = I$$

$$\langle \psi(t) | \psi(t) \rangle = U^\dagger(t) \cdot U(t) \cdot \langle \psi(0) | \psi(0) \rangle$$

→ Spectral Decomposition

$$\hat{H} = \sum_n E_n |E_n\rangle \langle E_n| \quad |\psi(0)\rangle = \sum_n c_n |E_n\rangle$$

$$c_n = \langle E_n | \psi(0) \rangle$$

$$|\psi(t)\rangle = e^{-i\hat{H}t/\hbar} \sum_n c_n |E_n\rangle$$

$$e^{-i\hat{H}t/\hbar} |E_n\rangle = e^{-iE_n t/\hbar} |E_n\rangle$$

$$|\psi(t)\rangle = \sum_n c_n e^{-iE_n t/\hbar} |E_n\rangle$$

↓

$\hat{H}$ doesn't change coefficients $\{c_n\}$,
each energy eigenstate simply picks up a phase $e^{-iE_n t/\hbar}$

$$|\psi(0)\rangle = \frac{|E_1\rangle + |E_2\rangle}{\sqrt{2}} \rightarrow |\psi(t)\rangle = \frac{e^{-iE_1 t/\hbar} |E_1\rangle + e^{-iE_2 t/\hbar} |E_2\rangle}{\sqrt{2}}$$

$$\text{Relative Phase} \rightarrow e^{-i\Delta E t/\hbar} \quad \Delta E = E_2 - E_1$$

$$\hat{H}' = \hat{H} + \hat{C} \quad \hat{C} \rightarrow \text{constant}$$

$$\hat{H}' \equiv \hat{H} \rightarrow e^{-iEt/\hbar} \rightarrow \text{Global phase}$$

$$U(t) = \sum_n e^{-iE_n t/\hbar} |E_n\rangle \langle E_n|$$

Quantum Gates from Hamiltonians

X Gate

Rotates Bloch vector around x-axis with frequency ω

$$\hat{H} = \frac{\hbar \omega}{2} X \quad U(t) = e^{-i\omega X t/2} = \cos\left(\frac{\omega t}{2}\right) I - i \sin\left(\frac{\omega t}{2}\right) X$$

At $t = \frac{\pi}{\omega} \rightarrow U(t) = -iX \equiv X$

↳ We evolve the system

Y Gate

$$\hat{H} = \frac{\hbar \omega}{2} Y \quad t = \frac{\pi}{\omega} \quad U(t) = -iY \equiv Y$$

Z Gate

$$\hat{H} = \frac{\hbar \omega}{2} Z \quad t = \frac{\pi}{\omega} \quad U(t) = -iZ \equiv Z$$

Hamiltonian for e^- in magnetic field

$$U = -\mu \cdot B \quad \mu = \gamma S$$

$$U = -\gamma S \cdot B$$

$$B = (B_x, 0, 0)$$

$$S_i = \frac{\hbar}{2} \sigma_i$$

$$\hat{H} = -\frac{\hbar \gamma B_x}{2} X \quad \omega = -\gamma B_x$$

In general, $\hat{H} = \frac{\hbar \Omega}{2} (\vec{n} \cdot \vec{\sigma}) \quad \Omega = \gamma \sqrt{B_x^2 + B_y^2 + B_z^2}$

$$\vec{n} = \frac{\vec{B}}{|\vec{B}|}$$

→ Time Independent Hamiltonian

$$\text{Total evolution} \rightarrow e^{-iH(t_1) \cdot \Delta t / \hbar} \cdot e^{-iH(t_2) \cdot \Delta t / \hbar} \cdot e^{-iH(t_3) \cdot \Delta t / \hbar}$$

$$= U(t) = \mathcal{T} \cdot e^{-i\hat{H}' \cdot t / \hbar}$$

$$\hat{H}' = \int_0^t \hat{H}(t) \cdot dt$$

$\mathcal{T} \rightarrow$ Time ordering operation

needed as $[\hat{H}(t_i), \hat{H}(t_j)] \neq 0$

$\mathcal{T} = I$ if and only if $[\hat{H}(t_i), \hat{H}(t_j)] = 0$
for all $\{i, j : t_i, t_j \in [0, t]\}$

→ Dyson series

$$U(t) = I + \left(\frac{-i}{\hbar}\right) \int_0^t H(t_1) \cdot dt_1 + \left(\frac{-i}{\hbar}\right)^2 \int_0^t dt_1 \int_0^{t_1} H(t_1) \cdot H(t_2) \cdot dt_2$$

$$U(t, t_0) = \sum_{n=0}^{\infty} \frac{1}{n!} \left(\frac{-i}{\hbar}\right)^n \int_{t_0}^t dt_n \int_{t_0}^{t_n} dt_{n-1} \dots \int_{t_0}^{t_2} dt_1 \mathcal{T}(H(t_n) \dots H(t_1))$$

→ von Neumann Equation

$$i\hbar \frac{\partial \rho}{\partial t} = [\hat{H}, \rho]$$

$$\begin{aligned} \frac{\partial \rho}{\partial t} &= \frac{\partial | \psi \rangle \langle \psi |}{\partial t} + | \psi \rangle \cdot \frac{\partial \langle \psi |}{\partial t} - \rho = | \psi \rangle \langle \psi | \\ &= \left(-\frac{i}{\hbar} \hat{H} | \psi \rangle \right) \langle \psi | + | \psi \rangle \left(\frac{i}{\hbar} \langle \psi | \hat{H} \right) \\ &= -\frac{i}{\hbar} (\hat{H} \rho - \rho \hat{H}) \end{aligned}$$

$$\frac{\partial \rho}{\partial t} = -\frac{i}{\hbar} [\hat{H}, \rho] \quad \text{with } \frac{\partial \rho}{\partial t} = [\hat{H}, \rho]$$

$$\hat{H} | E \rangle = E | E \rangle \quad - [\hat{H}, \rho] = 0 \quad - \frac{\partial \rho}{\partial t} = 0$$

$$\rho(t) = e^{-iEt/\hbar} | E \rangle \langle E | e^{iEt/\hbar} = | E \rangle \langle E |$$

↓
Considering environment $\rightarrow \rho_s = \text{Tr}_e(\rho)$

↓
Master equation, Markovian
and non-Markovian evolution

↓
Lindblad Master Equation